

GKE G4 Dual-NIC High-Bandwidth Setup & Benchmarking Guide

Version: 11.1

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Hardware Target: G4 Instances (NVIDIA RTX PRO 6000, 2x 200 Gbps Diorite SmartNICs)

Part 1: Cluster & Infrastructure Setup

1. Create the High-Bandwidth DRANET VPC & Subnets

Shell

```
# Create the Dedicated 8896 MTU VPC
gcloud compute networks create dranet-v11-dranet-vpc --subnet-mode=custom
--mtu=8896 --project=[PROJECT_NAME]

# Create Subnets for Control (0) and Payloads (1 & 2)
gcloud compute networks subnets create chavoshi-v11-sub0
--network=chavoshi-v11-dranet-vpc --range=10.10.0.0/16 --region=us-central1
--project=[PROJECT_NAME]
gcloud compute networks subnets create chavoshi-v11-sub1
--network=chavoshi-v11-dranet-vpc --range=10.20.0.0/16 --region=us-central1
--project=[PROJECT_NAME]
gcloud compute networks subnets create chavoshi-v11-sub2
--network=dranet-v11-dranet-vpc --range=10.30.0.0/16 --region=us-central1
--project=[PROJECT_NAME]

# Open Internal VPC Firewall Ingress
gcloud compute firewall-rules create allow-dranet-internal \
--network=chavoshi-v11-dranet-vpc --allow=tcp,udp,icmp
--source-ranges=10.0.0.0/8 --project=[PROJECT_NAME]
```

2. Provision the GKE Cluster & DRANET Node Pool

Shell

```
# Create the Cluster
gcloud container clusters create chavoshi-v11-cluster \
--network=chavoshi-v11-dranet-vpc --subnetwork=chavoshi-v11-sub0
--enable-ip-alias --enable-dataplane-v2 \
--zone=us-central1-b --num-nodes=1 --machine-type=e2-standard-4
--project=[PROJECT_NAME]
```

Create the Node Pool

```
gcloud container node-pools create chavoshi-v11-pool \
  --cluster=chavoshi-v11-cluster --machine-type=g4-standard-384 \
  --accelerator type=nvidia-rtx-pro-6000,count=8 --enable-gvnic --spot
--num-nodes=2 --zone=us-central1-b \
  --node-labels=cloud.google.com/gke-networking-dra-driver=true \
  --additional-node-network
network=chavoshi-v11-dranet-vpc,subnetwork=chavoshi-v11-sub1 \
  --additional-node-network
network=dranet-v11-dranet-vpc,subnetwork=chavoshi-v11-sub2
--project=[PROJECT_NAME]
```

3. Deploy Dynamic Resource Allocation Templates

Save as `netdev-template.yaml`:

```
None
apiVersion: resource.k8s.io/v1
kind: ResourceClaimTemplate
metadata:
  name: netdev-claim-template
spec:
  spec:
    devices:
      requests:
      - name: netdev-request
        exactly:
          deviceClassName: netdev.google.com
          allocationMode: ExactCount
          count: 2
```

```
Shell
kubectl apply -f netdev-template.yaml
```

4. Deploy the 32-Stream Workload Manifest

Save as `nccl-workload.yaml`:

```
None
apiVersion: v1
kind: ConfigMap
metadata:
```

```

    name: nccl-golden-script
data:
  dist_test.py: |
    import torch, os, time
    import torch.distributed as dist
    def main():
        os.environ["NCCL_SOCKET_IFNAME"] = "eth1,eth2"
        os.environ["NCCL_SOCKET_FAMILY"] = "AF_INET"
        os.environ["NCCL_DEBUG"] = "INFO"
        os.environ["NCCL_CROSS_NIC"] = "1"
        dist.init_process_group(backend="nccl")
        rank = dist.get_rank()
        device = torch.device(f"cuda:{rank % torch.cuda.device_count()}")
        elements = 128 * 1024 * 1024
        tensor = torch.ones(elements, dtype=torch.float32, device=device) * (rank
+ 1)

        for _ in range(5):
            dist.all_reduce(tensor)
            torch.cuda.synchronize()
        start = time.time()
        for _ in range(30):
            dist.all_reduce(tensor)
            torch.cuda.synchronize()
        avg_time = ((time.time() - start) / 30) * 1e6
        bytes_transferred = elements * 4 * 2
        algbw = (bytes_transferred / (avg_time / 1e6)) / 1e9
        if rank == 0:
            print(f"=== TRUE CROSS-NODE NCCL ALLREDUCE RESULTS ===")
            print(f"Payload Size: {elements * 4 / 1024 / 1024} MB")
            print(f"Duration per AllReduce: {avg_time:.2f} us")
            print(f"Measured Aggregate AlgBW: {algbw:.2f} GB/s")
            print(f"=====")
        dist.destroy_process_group()
    if __name__ == "__main__":
        main()
---
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nccl-afinet
  labels:
    app: nccl-afinet
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nccl-afinet
  template:

```

```
metadata:
  labels:
    app: nccl-afinet
spec:
  tolerations:
    - key: "nvidia.com/gpu"
      operator: "Exists"
      effect: "NoSchedule"
  affinity:
    nodeAffinity:
      requiredDuringSchedulingIgnoredDuringExecution:
        nodeSelectorTerms:
          - matchExpressions:
              - key: cloud.google.com/gke-nodepool
                operator: In
                values:
                  - chavoshi-v11-pool
    podAntiAffinity:
      requiredDuringSchedulingIgnoredDuringExecution:
        - labelSelector:
            matchExpressions:
              - key: app
                operator: In
                values:
                  - nccl-afinet
          topologyKey: "kubernetes.io/hostname"
  containers:
    - name: nccl-chamber
      image: docker.io/lmsysorg/sglang:v0.5.12-cu130
      env:
        - name: NCCL_SOCKET_FAMILY
          value: "AF_INET"
        - name: NCCL_IB_DISABLE
          value: "1"
        - name: NCCL_MIN_NCHANNELS
          value: "8"
        - name: NCCL_SOCKET_NTHREADS
          value: "8"
        - name: NCCL_NSOCKS_PERTHREAD
          value: "4"
        - name: NCCL_P2P_LEVEL
          value: "SYS"
        - name: NCCL_SOCKET_RCVBUF
          value: "8388608"
        - name: NCCL_SOCKET_SNDBUF
          value: "8388608"
        - name: NCCL_BUFFSIZE
          value: "16777216"
```

```

- name: NCCL_PROTO
  value: "Simple"
securityContext:
  privileged: true
  capabilities:
    add: ["NET_ADMIN", "IPC_LOCK", "SYS_PTRACE"]
command: ["/bin/bash", "-c"]
args:
- |
  apt update && apt install -y iproute2 net-tools iperf3
  ip link set eth1 mtu 8896
  ip link set eth2 mtu 8896
  sleep 3600
resources:
  limits:
    nvidia.com/gpu: "1"
  requests:
    nvidia.com/gpu: "1"
  claims:
    - name: netdev-claim
volumeMounts:
- name: script-volume
  mountPath: /root
- name: dshm
  mountPath: /dev/shm
resourceClaims:
- name: netdev-claim
  resourceClaimTemplateName: netdev-claim-template
volumes:
- name: script-volume
  configMap:
    name: nccl-golden-script
- name: dshm
  hostPath:
    path: /dev/shm

```

Shell

```
kubectl apply -f nccl-workload.yaml
```

Part 2: Benchmarking Validations & Exact Commands

1. Iperf3 8-Way Parallel Aggregate Throughput Test

Step 1.1: Get Pod Names & Pod IPs

```
Shell
kubectl get pods -l app=nccl-afinet -o wide
```

Expected Output:

```
None
NAME                                READY   STATUS    RESTARTS   AGE   IP
NODE
nccl-afinet-66f4c98d7d-6tqj8      1/1     Running   0           45s   10.60.2.8
gke-chavoshi-v11-clu-chavoshi-v11-poo-9a094df5-h81d
nccl-afinet-66f4c98d7d-8lgsb      1/1     Running   0           45s   10.60.1.8
gke-chavoshi-v11-clu-chavoshi-v11-nzcb
```

Step 1.2: Retrieve Target Pod eth1 IP address

```
Shell
kubectl exec nccl-afinet-66f4c98d7d-8lgsb -- ip addr show eth1
```

Expected Output:

```
None
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 8896 qdisc mq state UP group
default qlen 1000
    inet 10.30.0.2/32 scope global eth1
```

Step 1.3: Start the Iperf3 Server on Pod 1

```
Shell
kubectl exec nccl-afinet-66f4c98d7d-8lgsb -- iperf3 -s -D
```

Step 1.4: Execute the 8-way Parallel Stream from Pod 2 targeting eth1

```
Shell
kubectl exec nccl-afinet-66f4c98d7d-6tqj8 -- iperf3 -c 10.30.0.2 -P 8 -t 5
```

Expected Output Log:

```
None
Connecting to host 10.30.0.2, port 5201
[ ID] Interval          Transfer      Bitrate      Retr
[SUM]  0.00-5.00    sec    111 GBytes    191 Gbits/sec      0
[SUM]  0.00-5.00    sec    111 GBytes    191 Gbits/sec
sender
receiver

iperf Done.
```

2. NCCL AllReduce Cross-Node Test

Step 2.1: Start torchrun Rank 1 (Worker Node Pod 2)

```
Shell
kubectl exec nccl-afinet-66f4c98d7d-6tqj8 -- torchrun --nproc_per_node=1
--nnodes=2 --node_rank=1 --master_addr=10.60.1.8 --master_port=29540
/root/dist_test.py
```

Step 2.2: Start torchrun Rank 0 (Master Node Pod 1)

```
Shell
kubectl exec nccl-afinet-66f4c98d7d-8lgsb -- torchrun --nproc_per_node=1
--nnodes=2 --node_rank=0 --master_addr=10.60.1.8 --master_port=29540
/root/dist_test.py
```

Expected Output Log:

```
None
=== TRUE CROSS-NODE NCCL ALLREDUCE RESULTS ===
Payload Size: 512.0 MB
Duration per AllReduce: 28849.45 us
Measured Aggregate AlgBW: 37.22 GB/s
```

